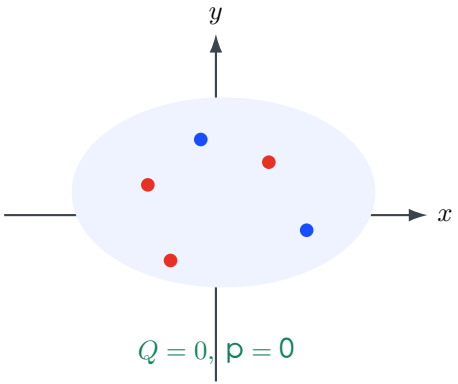
四极矩的原点不变性：必须同时满足总电荷与偶极矩为零



$$\mathbf{r}'_a = \mathbf{r}_a - \mathbf{a}$$

$$\mathcal{Q}'_{ij} = \mathcal{Q}_{ij} - 3(a_i p_j + a_j p_i) + 2\delta_{ij}\mathbf{a}\cdot\mathbf{p} + (3a_i a_j - a^2\delta_{ij})Q$$

$$Q = 0,\ \mathbf{p} = \mathbf{0} \implies \mathcal{Q}'_{ij} = \mathcal{Q}_{ij}$$